

Reproducible Values, Unstable Conclusions: An Interpretation Audit of Security Simulation

Anonymous authors

Abstract—Reproducing simulation numbers does not guarantee that a paper interprets them correctly. We audit one synthetic hospital ransomware study whose software checks and generated manuscript values coexisted with incorrect scope and uncertainty claims. Without changing the simulator or its 160,000 confirmation executions, we reanalyze paired scenarios, preserve the fixed allocation across attack-entry categories, and check selected interpretations with explicit executable assertions. Changing the required outage-service count shifts the high-capacity profile’s candidate-average event frequency by 14.34 percentage points while leaving its estimated six-objective Pareto frontier unchanged. A frozen discovery shortlist omits 38 efficient candidates in the observed bank; the conditional stratified-bootstrap range is 29–49. Although the observed shortlist contains no restricted-view artifacts, such artifacts appear in 42.7% of resamples. Covariance-aware uncertainty also contradicts a blanket interpretation of no detectable control benefit. Our contribution is a retrospective security-experimentation case study linking checkable claims to population, pairing, and decision-set diagnostics. It provides neither a general natural-language validator nor validated hospital recommendations.

Index Terms—security experimentation; simulation; interpretation audit; paired analysis; Pareto frontier; reproducibility

1. Introduction

Security simulations support comparisons that are difficult to observe in operational environments: what would the same attack have done under a different defense? Their usefulness depends on more than executing code or reproducing a table. A sentence can copy a correct value while changing its population, uncertainty interpretation, or decision meaning. These errors can survive both software tests and automated manuscript checks.

We investigate this gap through a retrospective audit of one open synthetic hospital ransomware study. The preceding version already included paired scenarios, frozen protocols, generated numeric macros, archived evidence, mechanism tests, and explicit limitations. Its manuscript nevertheless described large profile-specific changes as uniformly small, mislabeled a figure’s experimental stage, and interpreted a heuristic separation rule as an inability to detect differences. Reproducing these contradictions is more

informative than treating successful execution as the end of verification.

The setting matters because plausible simulation conclusions can be mistaken for healthcare guidance. Empirical incident research documents disruption, but does not identify the counterfactual effect of each proposed control [1]. Process-level hospital simulation can incorporate local workflow observations and expert input [2]; the model studied here does not have that level of validation. We use it as an inspectable security-experimentation case, not as a validated procurement instrument.

Our question is: *which conclusions change when interpretation is checked at the same level of specificity as the computation?* We contribute three case-specific results. First, a substantial change in an outcome frequency can coexist with an unchanged estimated Pareto frontier. Second, retaining scenario pairing and the fixed entry-category design changes uncertainty diagnostics; stable rankings and lack of evidence for a difference are separate claims. Third, the observed completeness of a discovery shortlist depends on conditional resampling and on declared objective weights. A small executable assertion registry makes selected scope, unit, aggregation, and bound claims inspectable alongside these results.

We do not claim to introduce preregistration, common random numbers, metamorphic testing, or structured claim auditing. The contribution is a documented failure-and-repair case with quantitative interpretation diagnostics. The new analysis is retrospective and informed by prior reviews; it is not a new preregistered confirmation or an independent replication.

2. Study and Audit Design

2.1. The preserved simulation experiment

The simulator represents one synthetic facility as a network of digital assets with clinical-service dependencies. An incident enters through one of five declared categories: workstation, internet-facing asset, privileged system, medical device, or vendor connection. Spread, detection, defensive isolation, and restoration evolve in five-minute steps over a 72-hour horizon. Clinical service availability is binary at each step, using a declared 60% supporting-node threshold. This is a digital-service construct, not a measurement of patients treated, clinical harm, or full organizational recovery.

Three capacity profiles are fixed environments. Controls can change segmentation, patching, detection, isolation, backups, and identity protection; they cannot purchase a change of profile. Patch protection is mechanism specific, identity protection applies to covered authentication pathways, and isolated backups can still suffer a per-incident isolation lapse. The revision preserves these mechanisms, all original raw trials, and the historical protocols. Full model details and earlier sensitivity experiments remain in the archived manuscript and repository.

There are 288, 96, and 16 distinct resolved candidates in the resource-constrained, intermediate, and high-capacity profiles. Discovery contains 30 scenarios per profile and 12,000 candidate executions. Confirmation contains 400 fresh scenarios per profile and 160,000 candidate executions. Scenario IDs occupy disjoint blocks across profiles and stages. Within a profile, each scenario is shared across all candidates using event-keyed random inputs. The confirmation design cycles through the five entry categories, giving 80 scenarios per category. Thus 160,000 executions do not constitute 160,000 independent experimental units. The estimand averages equally across the five designed categories, not across an empirically estimated hospital-incident population.

Each candidate has six objectives, all minimized: mean weighted service-hours lost, empirical upper-tail mean loss, sustained-outage frequency, non-recovery by 72 hours, implementation cost, and operational burden. The historical tail statistic is the mean at or above the empirical 90th percentile, including all ties; we retain that convention rather than silently replacing it with a different expected-shortfall estimator. Cost and burden are normalized scenario points, not dollars or measured staff effort. Technical control effects remain non-identifiable from the available public sources; the model is only partially calibrated in its context and incident constraints.

2.2. Outcome definitions and population alignment

Let N_{pci} be the number of clinical services with an individual continuous outage longer than two hours in profile p , candidate c , scenario i . The event $Y_{pci}^{(k)} = \mathbf{1}\{N_{pci} \geq k\}$ requires at least k qualifying services; their outage intervals need not overlap. The primary endpoint uses $k = 4$. We report all four values of k and recompute the candidate-average frequency at each $k \in \{1, 2, 3, 4\}$ separately for each stage and profile. Averaging over candidates describes this enumerated design, not an incident prevalence.

For the same observations and weights, the following identity holds exactly:

$$\overline{Y^{(1)}} - \overline{Y^{(4)}} = \overline{\mathbf{1}\{1 \leq N \leq 3\}}. \quad (1)$$

The fraction of partial-service outcomes is therefore the endpoint's $k = 1$ versus $k = 4$ sensitivity, not independent evidence for its insensitivity. The identity does not justify moving a pooled discovery statistic into a profile-specific confirmation claim. We check the identity numerically and

report both candidate-weighted pooling and equal-profile averaging.

2.3. Pairing, allocation, and uncertainty

For two candidates, define the paired loss difference D_{hi} in entry stratum h , with allocation weight $w_h = n_h/n$. The estimated mean-difference variance is

$$\widehat{\text{Var}}(\widehat{\Delta}) = \sum_{h=1}^5 w_h^2 \frac{s_{D,h}^2}{n_h}, \quad s_{D,h}^2 = s_{a,h}^2 + s_{b,h}^2 - 2s_{ab,h}. \quad (2)$$

Dropping covariance changes the question's uncertainty. It can either inflate or deflate an estimate depending on covariance; pairing is not guaranteed to help for every possible model. For the nonlinear tail statistic, covariance is estimated from shared stratified bootstrap draws.

We compare paired and covariance-omitting versions of the earlier "within one standard error on all four stochastic objectives" diagnostic, using the same design and bootstrap draws. This is a descriptive heuristic, not a hypothesis test, equivalence test, or proof that portfolios can be ordered. For a more explicit mean-loss comparison, every nonfree candidate is compared with its own profile's free baseline. Approximate paired t intervals use Satterthwaite degrees of freedom and Bonferroni adjustment across all 397 contrasts. This includes the observed best candidate in the family rather than presenting it as a prespecified isolated comparison. Zero observed within-stratum variance is marked unestimable, not certain.

2.4. Decision-set and specification diagnostics

A candidate is Pareto efficient if no other candidate is no worse on every included objective and strictly better on at least one. We enumerate the full finite space; no optimization heuristic approximates this set. Let $F(X)$ denote the frontier in candidate set X , and let S be the 65 candidates labeled finalists by the frozen discovery rule. We distinguish *omissions*, $F(X) \setminus S$, from *restricted-view artifacts*, $F(S) \setminus F(X)$. A shortlist can omit efficient candidates without containing any spurious efficient candidate itself.

We resample whole scenarios within each entry category, preserving all candidate and endpoint pairings. Each of 1000 draws recomputes the six objectives and both frontiers. Independent profile blocks are resampled separately. Finalist labels remain fixed: the analysis conditions on the observed discovery decision and does not rerun its selection rule. Percentile summaries are conditional resampling ranges, not confidence sets for a uniquely true frontier or forecasts for other hospitals. Discrete dominance and selection make ordinary bootstrap coverage especially uncertain.

We also enumerate all 63 nonempty subsets of the six objectives and compare frontier membership across k . Jaccard similarity is intersection size divided by union size. Finally, 500 stress draws independently multiply each of eight cost and eight burden coefficients by a $\text{Uniform}(0.5, 1.5)$ factor, shared across profiles. Draws reversing the cumulative

TABLE 1. CONFIRMATION CANDIDATE-AVERAGE EVENT FREQUENCIES (%), THE $k = 1$ TO $k = 4$ DIFFERENCE (PERCENTAGE POINTS), AND FRONTIER JACCARD SIMILARITY.

Profile	$k = 1$	$k = 4$	Gap	Jaccard
Resource-constrained	87.32	83.45	3.87	0.896
Intermediate	57.39	44.79	12.60	1.000
High-capacity	16.00	1.66	14.34	1.000

order of the segmentation or backup tariffs are rejected (186 rejections). These are analyst-declared perturbations, not empirical priors. Positive uniform scaling of cost and burden is checked separately: it preserves Pareto dominance algebraically, although keeping a numeric budget fixed can change affordability.

2.5. Executable interpretation assertions

The registry gives each selected claim a source table, scope filter, expected groups and row count, statistic, unit, reduction, quantifier, relation, and bound. For example, interpreting a claim that all profile gaps are roughly three percentage points as $\max_p \Delta_p \leq 4$ percentage points requires the confirmation population and every profile. Four is a generous audit tolerance for that wording, not a clinical cutoff. Failed assertions return group-specific witness values. Missing populations, unit mismatches, and unsupported operations are invalid and stop verification.

The finite evaluation has 8 correct assertions, 3 historical contradictions, and 8 single-field mutations of correct assertions. It includes exact-boundary, missing-population, and unsupported-operation cases. These were constructed after reviewing the paper. Detection on this set demonstrates the specified checks, not sensitivity or specificity on unseen manuscripts. A human must still judge whether an assertion captures the intended sentence and whether the source itself has scientific validity.

3. Results

3.1. Endpoint frequency and frontier stability separate

Table 1 rejects the historical claim of uniformly small k sensitivity. The stronger two profiles have double-digit percentage-point changes, yet their frontier membership is identical across all four values of k . In contrast, the resource-constrained frontier changes membership despite a smaller frequency change. An unchanged frontier means dominance relations survive this specification change; it does not mean the event being reported occurs at approximately the same frequency.

Candidate-weighted pooling compounds the problem. Resource-constrained candidates contribute 72% of executions, intermediate candidates 24%, and high-capacity candidates 4%. In discovery, the fraction with one to three

TABLE 2. PAIRS MEETING THE ONE-SE PROXIMITY HEURISTIC ON ALL STOCHASTIC OBJECTIVES, WITH IDENTICAL STRATIFICATION AND TAIL-BOOTSTRAP DRAWS. “OMITTED” DROPS COVARIANCE; “PAIRED” RETAINS IT. NEITHER COLUMN IS A COUNT OF STATISTICALLY EQUIVALENT PAIRS.

Profile	All pairs	Omitted	Paired
Resource-constrained	41,328	770	88
Intermediate	4,560	84	10
High-capacity	120	32	2

TABLE 3. OBSERVED SIX-OBJECTIVE FRONTIERS. RESTRICTED EVALUATES ONLY FROZEN FINALISTS. OMISSIONS ARE FULL-SPACE EFFICIENT CANDIDATES OUTSIDE THE SHORTLIST; ARTIFACTS ARE EFFICIENT ONLY WITHIN THE SHORTLIST.

Profile	Full	Restricted	Omitted	Artifacts
Resource-constrained	64	36	28	0
Intermediate	22	17	5	0
High-capacity	9	4	5	0

qualifying services is 3.10% pooled but 6.46% with equal profile weights; the intermediate and high-capacity values are 9.03% and 9.58%. In confirmation, the corresponding pooled and equal-profile values are 6.38% and 10.27%. Neither weighting is an empirical hospital mixture. The original discovery aggregate could therefore support neither a universal profile claim nor a claim about the confirmation stage. Figure 1 and Eq. 1 make both distinctions visible.

3.2. Pairing changes the uncertainty interpretation

The comparison in Table 2 isolates covariance within the corrected design. It differs from comparing against historical published counts, which also used different resampling draws and did not preserve the fixed entry allocation. The large reduction in near-pair counts shows why calling this heuristic “indistinguishability” was misleading. We do not interpret pairs outside the heuristic as statistically ordered.

The high-capacity candidate with the lowest observed mean loss saves 1.95 weighted service-hours against its free baseline. Its stratified paired SE is 0.333, compared with 0.944 when covariance is omitted. The approximate simultaneous interval across all 397 baseline contrasts is [0.66, 3.25] hours. Thus the bank supports a positive within-model mean difference even under a family-wide adjustment. It does not establish clinical importance, affordability, or transportability. It also contradicts the earlier figure caption describing a vertical range of less than one hour: the baseline-to-best difference alone exceeds that.

3.3. A single observed frontier is incomplete evidence

The full-space frontier has 95 candidates, compared with 57 in the restricted view; the latter omits 38 full-space efficient candidates (Table 3). Across the stratified

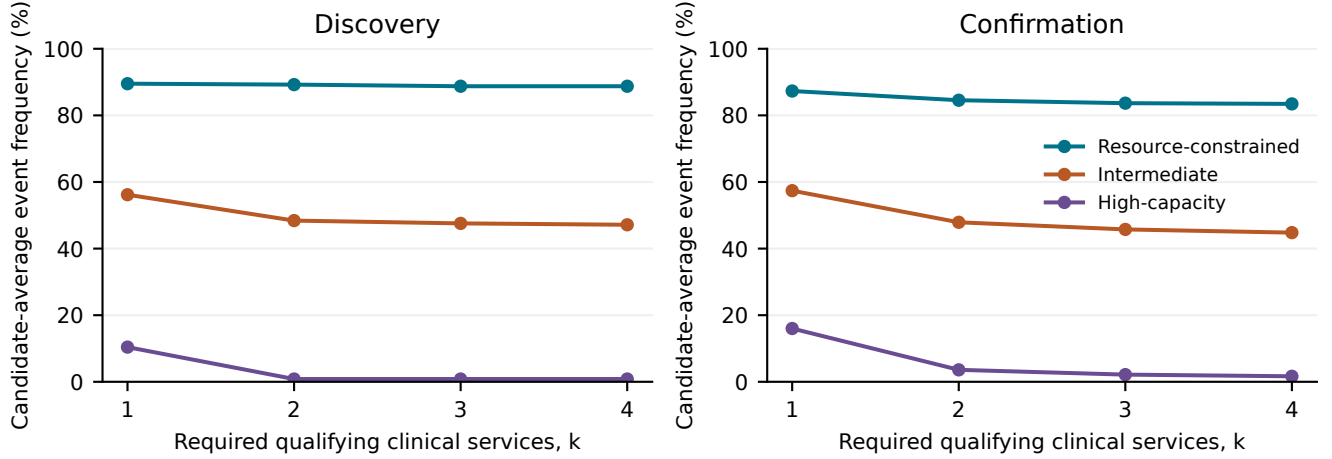


Figure 1. Separate discovery and confirmation distributions. Each point averages over all resolved candidates in one profile. The vertical axis describes digital-service events under the synthetic experiment, not real hospital incident prevalence. Connecting the nested thresholds shows their frequency sensitivity; it does not imply interpolation between integer k .

TABLE 4. FRONTIER SENSITIVITY TO 500 DECLARED COMPONENT-WEIGHT STRESS DRAWS. SIMILARITY IS RELATIVE TO THE ORIGINAL SIX-OBJECTIVE FRONTIER; ADDED AND REMOVED COLUMNS ARE MAXIMA ACROSS DRAWS.

Profile	Median J	Min J	Added	Removed
Resource-constrained	0.728	0.500	58	14
Intermediate	0.724	0.438	26	8
High-capacity	0.818	0.500	6	3

resamples, the rounded central 95% ranges are 83–106 for the full frontier and 29–49 for omissions. Every resample contains an omission. The existence of omissions is stable under this conditional exercise; the exact count is not.

Zero restricted-view artifacts in the observed bank does not imply a reliably artifact-free shortlist. At least one appears in 42.7% of conditional resamples (Figure 2). This percentage is not the probability that an independently selected shortlist would fail: discovery is not resampled. The distinction prevents a reassuring zero from being generalized beyond the evidence that produced it.

3.4. Objective and tariff choices matter

Heterogeneous tariff changes alter frontier membership in every profile (Table 4). A uniform conversion of units preserves dominance, so varying a single common multiplier would have missed this dependence. The result identifies sensitivity to declared assumptions; the chosen perturbation interval does not supply an empirical distribution of costs.

The objective ablation also cautions against treating the total frontier size as surprising by itself. Across two-objective subsets, frontier size ranges from 1–16, 1–18, and 1–9 in the three profiles. Across all six objectives it is 64, 22, and 9. However, tied single-objective minima can be numerous: one intermediate objective has 47 tied efficient

candidates. The appendix reports every dimension. Frontier size depends on objective choice, dependence, and ties; this study does not provide an independent-objective null model that predicts 95 efficient candidates.

3.5. What the assertion audit detects

The preserved number/regex audit passes on the preserved manuscript despite the three registered contradictions: the universal small-gap claim, the discovery label on a confirmation figure, and the under-one-hour range. The new assertions reject all 3 with witness values, accept the 8 correct cases, and reject or flag invalid all 8 controlled mutations as specified. The latter include a wrong stage, wrong unit, incomplete population, wrong aggregation, wrong bound, strict-versus-inclusive boundary, missing population, and unsupported operation. This is a finite regression demonstration. It does not show that adding more assertions will discover an unknown omitted claim or that a human translation of prose is correct.

4. Relation to Prior Work

Simulation reporting guidelines, especially STRESS, already call for transparent model descriptions, experimental design, and implementation details [3]. Preregistration distinguishes prior commitments from outcome-informed analysis [4]. Our retrospective plan and amendment use that distinction explicitly; recording them does not transform an audit informed by observed defects into a confirmatory experiment.

Common random numbers are an established variance-reduction strategy. Event-keyed random inputs can preserve meaningful coupling when alternative policies change execution order [5]. This study does not introduce that technique. Its additional observation is that a paired simulator can still

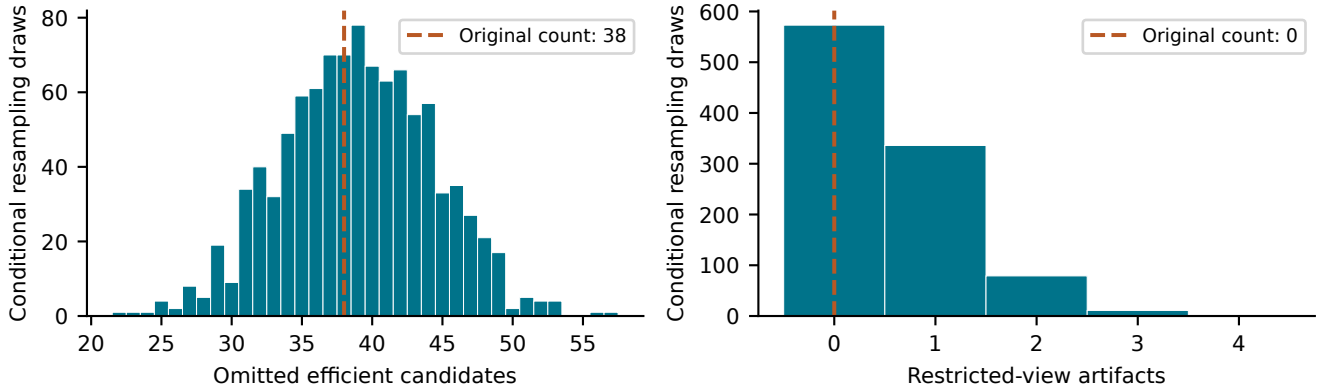


Figure 2. 1000 stratified scenario resamples, summed across the three independent profile banks. Finalist labels are fixed. Dashed lines show original-bank counts. These are conditional resampling distributions, not confidence intervals for a true frontier or new hospital population.

be followed by an uncertainty calculation that discards the resulting covariance. The analysis must also retain fixed design strata; correct pairing alone is insufficient.

Multiverse and specification-curve approaches expose variation across defensible analytical choices [6], [7]. Our endpoint, objective-subset, and tariff exercises are narrower descriptive diagnostics. We do not implement the joint inference of a full specification-curve analysis or claim that every enumerated specification has equal substantive merit. Their purpose is to separate endpoint-frequency sensitivity from decision-set sensitivity within one finite experiment.

Automated statistical consistency checks such as statcheck address whether reported test statistics and p-values agree [8]. Metamorphic testing examines expected relations when direct simulation oracles are unavailable [9]. Our endpoint identity and unit-scaling check are examples of this broader relational-testing tradition. The small claim registry adds explicit population, units, and quantifiers for selected interpretations; it does not replace either numerical or model validation.

A recent preprint, ABE-Ralph, represents experimental claims and protocols as structured constraints and checks fidelity in LLM-driven scientific workflows [10]. It is directly relevant prior work, so a broad claim to be the first executable scientific-claim auditor would be unwarranted. Our narrower focus is retrospective statistical interpretation of a security simulation, including paired covariance, fixed entry allocation, and conditional Pareto-set diagnostics. Unlike a general claim-extraction system, our evaluator only handles explicitly authored assertions. The assertions are useful because their assumptions and failure witnesses are small enough to inspect, not because they automate scientific judgment.

Finally, healthcare cyber-disruption models can address process overload using observations and expert assessment [2]. Our case lacks independent clinical validation and cannot compete on predictive realism. Its research value lies in exposing errors that could also occur in a more realistic model: a valid computation can still be summarized with

the wrong population or strength of conclusion.

5. Limitations, Ethics, and Transfer

5.1. What the evidence does not establish

This is one retrospectively selected project, not a representative sample of security papers. The reviewers' observations shaped both the analyses and the finite assertion cases. Multiple defects within this project are not independent replications of the method. Independent application to a second simulator and blind evaluation of assertions on unseen manuscripts would be stronger evidence of general usefulness.

The resampling preserves scenario pairing and the experiment's balanced entry categories, but conditions on the model, parameter distribution, observed bank, and frozen shortlist. It does not capture unmodeled recovery processes, uncertainty in the clinical-service construct, or the variability of choosing a different discovery bank. Pareto selection is discontinuous, especially with tied or nearly equal objectives. Reported percentile ranges are descriptive conditional diagnostics; no bootstrap coverage guarantee is established. The adjusted t intervals are approximate mean inference and may remain sensitive to tail behavior and finite within-stratum samples.

Earlier parameter and structural sensitivity experiments remain in the artifact. They cannot identify coefficients absent measured correspondences between model mechanisms and real controls. The new tariff draws likewise test a chosen range, not a validated uncertainty distribution. A beneficial simulated mean contrast does not by itself answer whether a hospital should buy a control or whether the improvement has clinical importance.

5.2. Validation failures remain visible

The historical external-validation registry has nine benchmarks. Two are numerically scored: one fails and

one is loosely consistent. Two additional benchmarks are recorded as direct qualitative failures; three are not scored and two are structurally unaddressable. Thus the total of three failures includes one numeric failure and two qualitative failures; it is not three failed numeric tests out of two. The loose consistency label is not a calibration success: a broad simulated profile range contains a survey estimate from a different population.

The single-facility, 72-hour horizon cannot represent regional propagation or months of institutional recovery. Binary digital-service availability cannot represent partial clinical capacity or patient outcomes. These are construct limitations that more Monte Carlo trials cannot repair. The correct conclusion is limited to the declared computational experiment.

5.3. Audit scope, responsibility, and ethics

The workflow verifies more than bytes, but less than the truth of a paper. An incorrect source schema or an overly weak assertion can pass every check. For that reason, the registry, historical witnesses, and analysis plan are part of the artifact, and human review of each translation is still required. Hash checking preserves evidence about a run; it cannot establish that a dirty historical tree matched an uncommitted implementation or prove a claimed chronology retroactively. Clean source capture in this revision is a prospective workflow repair, not a revision of those historical facts.

This reanalysis uses existing synthetic trials and public, aggregate source material. It introduces no patient records, participant recruitment, deployment against live systems, or new offensive experiments. We make no claim of institutional ethics approval. The principal foreseeable harm is overinterpretation as clinical or procurement advice, addressed by reporting failed benchmarks and keeping modeled outcomes distinct from patient harm. Author contribution statements and independent domain review must be completed by the researchers, not inferred from a successful automated run.

The transferable practice is concrete: when a paper claims that a result is stable, identify whether that means stable frequencies, estimates, uncertainty intervals, or decision-set membership. Declare the population, pairing and allocation, and aggregation before checking the assertion. Report the failures that remain outside the model's scope alongside the checks that pass.

6. Conclusion

The audited simulation reproduced its numeric outputs while supporting incorrect statements about their scope and uncertainty. Reanalysis separates large endpoint-frequency changes from unchanged frontiers, preserves paired and stratified uncertainty, and shows why one observed shortlist does not establish stable frontier completeness. Explicit assertions detect the selected historical contradictions while

exposing the assumptions they rely on. This case supports a limited but useful standard for security experiments: reproduce the numbers, then make the interpretation specific enough to test.

Artifact Availability

The artifact includes the original and revised manuscript sources, preserved trial banks, frozen protocols, an explicit retrospective analysis plan and sampling-design amendment, claim registry, all analysis tables, generated figures, and verification scripts. Existing model and historical analyses remain available. Raw confirmation data are stored losslessly compressed and restored by the supplied script. The identified public repository is withheld from this review copy. A separate anonymous artifact package still needs to be arranged before a double-blind submission.

References

- [1] H. T. Neprash, C. C. McGlave, D. A. Cross *et al.*, "Trends in ransomware attacks on US hospitals, clinics, and other health care delivery organizations, 2016–2021," *JAMA Health Forum*, vol. 3, no. 12, p. e224873, 2022.
- [2] M. Willing, S. Ebberts, C. Dresen, M. Czolbe, C. Saatjohann, and S. Schinzel, "Simulating the overload of medical processes due to system failures during a cyberattack," *BMC Medical Informatics and Decision Making*, 2025.
- [3] T. Monks, C. S. M. Currie, B. S. Onggo, S. Robinson, M. Kunc, and S. J. E. Taylor, "Strengthening the reporting of empirical simulation studies: Introducing the STRESS guidelines," *Journal of Simulation*, vol. 13, no. 1, pp. 55–67, 2019.
- [4] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018.
- [5] V. Buffalo, C. A. B. Pearson, and D. Klein, "Realizing common random numbers: Event-keyed hashing for causally valid stochastic models," 2026.
- [6] S. Steegen, F. Tuerlinckx, A. Gelman, and W. Vanpaemel, "Increasing transparency through a multiverse analysis," *Perspectives on Psychological Science*, vol. 11, no. 5, pp. 702–712, 2016.
- [7] U. Simonsohn, J. P. Simmons, and L. D. Nelson, "Specification curve analysis," *Nature Human Behaviour*, vol. 4, pp. 1208–1214, 2020.
- [8] M. B. Nuijten, C. H. J. Hartgerink, M. A. L. M. van Assen, S. Epskamp, and J. M. Wicherts, "The prevalence of statistical reporting errors in psychology (1985–2013)," *Behavior Research Methods*, vol. 48, pp. 1205–1226, 2016.
- [9] M. S. Raunak and M. Olsen, "Metamorphic testing on the continuum of verification and validation of simulation models," in *IEEE/ACM 6th International Workshop on Metamorphic Testing (MET)*, 2021.
- [10] L. Yu, X. Xu, Y. Zhou, S. He, and A. Pan, "Beyond execution: Auditing experimental fidelity in LLM-driven scientific research," 2026, preprint, version 1, August 27, 2026.

LLM Usage Statement

This revision used OpenAI Codex for literature discovery, analysis design, code drafting, automated checks, and manuscript editing. Reported numerical results were computed by local scripts from preserved trial data; language

model output was not used as a numerical observation. The assertion registry is deterministic and does not call an LLM. LLM assistance can introduce incorrect references, implementation errors, or overstatements; source verification and executable checks reduce but do not eliminate those risks. Human author review of these additions remains pending. This is a working draft, and this statement must be completed with verified author oversight before submission. Earlier editorial assistance must also be confirmed by the authors in the final disclosure.

Appendix A. Reproduction and Audit Boundary

The original raw banks are unchanged. New analysis starts from a committed, clean source state captured before output generation. Each manifest records the source commit, environment, input and output sizes, and SHA-256 hashes. The historical generated-number manifest records a dirty working tree; we preserve that fact. Hash agreement verifies artifact consistency but cannot retroactively prove that the historical code was committed before execution. A commit-attribution map resolves historical source identifiers after the repository’s metadata-only attribution correction.

For the final retrospective analysis, Python 3.12 and exact package versions are recorded in the manifest and revision environment file. The reproduction guide gives commands to restore the compressed bank, run the verification harness, regenerate analysis tables from a clean checkout, and generate numeric TeX and vector figures after committing those tables. The paper generator’s check mode recomputes the TeX and verifies its manifest without overwriting files. These checks certify specified artifact relations, not the correctness of every sentence.

The retrospective plan records an amendment: early diagnostic output treated entry categories as randomly mixed, but source inspection established a fixed allocation. Final results preserve that allocation. Initial outputs are retained in Git history rather than relabeled as the final design. For Eq. 2, set $v_h = w_h^2 s_{D,h}^2 / n_h$ and use approximate degrees of freedom $(\sum_h v_h)^2 / \sum_h [v_h^2 / (n_h - 1)]$. The interval is $\hat{\Delta} \pm t_{1-.05/(2.397), \nu} \sqrt{\sum_h v_h}$.

Appendix B. Objective-Subset Enumeration

TABLE 5. FRONTIER SIZE OVER ALL OBJECTIVE SUBSETS OF EACH DIMENSION. ENTRIES ARE MINIMUM/MEDIAN/MAXIMUM, INCLUDING TIED OPTIMA. R, I, AND H DENOTE RESOURCE-CONSTRAINED, INTERMEDIATE, AND HIGH-CAPACITY PROFILES.

Dimensions	R	I	H
1	1/1/3	1/1/47	1/1.5/8
2	1/6/16	1/3/18	1/2/9
3	1/18.5/45	1/14/22	1/4.5/9
4	6/45/53	1/21/22	1/6/9
5	28/54/64	14/22/22	5/9/9
6	64/64/64	22/22/22	9/9/9

The single-objective intermediate maximum is caused by tied optima. Adding an objective can break such ties and remove a previously efficient candidate. Therefore frontier cardinality need not increase monotonically with dimension in this discrete problem. The enumeration characterizes this candidate set; it is not a distribution-free prediction of frontier size.